

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of M. Sc. (Biotechnology/Microbiology/Biochemistry) Examination, May 2018

Course code & Name: MS 851 & Research Methodology

Date: 02/05/2018 Day: Wednesday Time: 10:00 A.M. To 10:30 A.M. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	Of the following, which laboratory biosafety level containment is most appropriate for research on aerosol transmitted agents which may cause serious or life threatening infections upon inhalation?		01
	(i) BSL-3 & BSL-1	(ii) LAF & BSL-2	
	(iii) BSL-4 & BSL-3	(iv) BSL-2 & BSL-1	
2.	If scientist A, has published in total 10 research papers, out of which one of the paper has received 500 citations and the other nine papers received only one citation each. What will be the <i>h</i> -index of scientist A?		01
	(i) 1	(ii) 10	
	(iii) 509	(iv) 2	
3.	If a biomedical scientist studies the potential risks of cardiovascular complications in a group of patients with type 2 diabetes mellitus for over several longer periods of time, such type of research can be classified as?		01
	(i) Cross-sectional research	(ii) One-time research	
	(iii) Behavioral research	(iv) Longitudinal research	

4.	The following committee has to be constituted by research establishments in India, to ensure compliance for ethical conduct of animal experimentation, in accordance with the CPCSEA nominations, guidelines and supervision.		01
	(i) IBSC	(ii) IAEC	
	(iii) IRB	(iv) INSA	
5.	Of the following, which research approach can be sub-classified into inferential, experimental and simulation approaches.		01
	(i) Qualitative approach	(ii) Unstructured approach	
	(iii) Flexible approach	(iv) Quantitative approach	
6.	The following is classified under the Category No. 2 of biomedical wastes?		01
	(i) Human anatomical waste	(ii) Animal waste	
	(iii) Waste sharps/needles	(iv) Discarded medicines & cytotoxic drugs	
7.	The following bibliometric measure is used to evaluate the importance or the rank of an academic journal by calculating the yearly average number of citations received by articles published in that journal.		01
	(i) ICMJE	(ii) <i>h</i> -index	
	(iii) Impact factor	(iv) Boolean operators	
8.	The structural aspects of a biological research laboratory, such as special air-ventilation patterns can be classified as _____ for laboratory biosafety.		01
	(i) Personal protective equipment	(ii) Secondary barriers	
	(iii) Primary barriers	(iv) All of them	
9.	The following is an example of primary scientific literature.		01
	(i) Review article	(ii) Encyclopedia	
	(iii) Textbook	(iv) Research paper	
10.	Intellectual Property (IP) is the name given to intangible property that arise from creations of the mind. Which of the following can be categorized as an IP?		01
	(i) Patents	(ii) Copyrights	
	(iii) Trademarks	(iv) All of them	
11.	Of the following types of plagiarism in scientific literature, “Paraphrasing” can be categorized as _____.		01
	(i) Plagiarism of Ideas	(ii) Self-Plagiarism	
	(iii) Plagiarism of Text	(iv) Mosaic plagiarism	
12.	The following P-value when observed in your experimental results would suggest that your results are statistically significant and that you can reject the null hypothesis.		01
	(i) $P = 0.5$	(ii) $P < 0.05$	
	(iii) $P > 0.15$	(iv) All of them	

13.	The following is NOT a reference management software.		01
	(i) Mendeley	(ii) BibteX	
	(iii) RefWorks	(iv) iThenticate	
14.	The following unique international identifier is assigned only to text-based monographic book publications and is not assigned to journals, periodicals and newspapers.		01
	(i) ISSN	(ii) DOI	
	(iii) ISBN	(iv) SCI	
15.	The following is a non-probability sampling method.		01
	(i) Simple random sampling	(ii) Systematic sampling	
	(iii) Stratified sampling	(iv) Quota sampling	
16.	The formulation of a _____ is the first and most important step in the research process, which greatly influences all the consequential steps and outcomes of a study.		01
	(i) Research hypothesis	(ii) Research design	
	(iii) Sample design	(iv) Research problem	
17.	If a scientist develops 3D printed artificial kidneys as a model to investigate new surgical procedures, such as nephrectomy, this type of research approach can be classified as _____.		01
	(i) <i>In vivo</i> research approach	(ii) Inferential approach	
	(iii) Simulation approach	(iv) Survey approach	
18.	Of the following, which attribute is essential for conducting a good research study.		01
	(i) Empiricism	(ii) Systematic planning	
	(iii) Replicability	(iv) All of them	
19.	The following is NOT a research funding agency to submit grant applications.		01
	(i) BRNS	(ii) CINAHL	
	(iii) SERB	(iv) DBT	
20.	USPTO allows one-year grace period after public disclosure of an invention, in contrast many other countries require fulfillment of the _____ criteria for patentability of an invention.		01
	(i) Industrial design	(ii) Counterclaim	
	(iii) Claim feasibility	(iv) Absolute novelty	

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Course code & Name: MS 851 & Research Methodology

Date: 02/05/2018 Day: Wednesday Time: 10:30 A.M. To 1:00 P.M. Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q – II Answer the following questions as directed (answer any four).		20
1.	Why is informed consent required for research on human subjects? Explain the elements and the process of informed consent.	05
2.	Given an account of various considerations involved in selection of a research problem.	05
3.	Describe the recommended primary and secondary barriers in various types of laboratory biosafety-levels for working with infectious agents.	05
4.	Discuss briefly: (i) Characteristics of good research & (ii) Research methodology <i>versus</i> research methods	05
5.	With the help of an appropriate example, explain the systematic method in formulation of a research problem.	05
SECTION - II		
Q-III Answer the following questions as directed (answer any six).		30
1.	Explain briefly the following types of scientific research: (i) Correlational Research (ii) Fundamental Research (iii) Empirical Research (iv) Exploratory Research (v) Action Research	05
2.	Explain the calculation of <i>h</i> -index, state its strengths and limitations.	05
3.	Classify and discuss the features of various publications in scientific journals.	05
4.	Explain the different forms of plagiarism in scientific literature.	05
5.	Explain the IMRAD format of an experimental research article.	05
6.	Describe the different categories of scientific literature.	05
7.	Explain the important considerations to made for writing a research project proposal.	05
8.	Give an outline and explain the various steps involved in the research process.	05